

DSA Test

Duration: 90 minutes
Total Questions: 5

Question 1 – The Butterfly Wings

A biologist is studying butterfly wing symmetry. She wants to visualize a butterfly's wing pattern on her terminal. The butterfly has n layers — each layer adds one more cell to each wing, and the wings mirror each other with a gap in between that shrinks as the layers grow.

At the widest point (layer n), both wings touch. The bottom half of the butterfly mirrors the top half exactly.

Input: A single integer n ($1 \leq n \leq 20$) — the number of layers in each wing.

Output: Print the butterfly pattern using `*` for wing cells and spaces for gaps.

```
Input:
4

Output:
*      *
**     **
***    ***
*****
*****
***    ***
**     **
*      *
```

Question 2 – The Ancient Temple Door

An archaeologist has discovered an ancient temple with a mysterious door. The door has an $n \times n$ grid of tiles, and to open it, the tiles must be arranged in a specific pattern: a solid border with a hollow diamond shape carved into the center.

The archaeologist needs your help to generate the correct tile arrangement. Filled tiles are represented by `*` and empty tiles by spaces.

Input: A single odd integer n ($3 \leq n \leq 51$) — the size of the grid.

Output: Print the $n \times n$ grid with the hollow diamond pattern.

```
Input:
7

Output:
*****
***  **
*_**
**__**
*___*
*___*
**__**
***  **
*_**
*****

(underscores shown for clarity – use spaces in actual output)
```

Question 3 – The Secret Spy Network

During a covert operation, a spy agency intercepted a list of n codenames. Intelligence suggests that agents who are anagrams of each other belong to the same cell. Your mission is to group all codenames that belong to the same cell together so the agency can identify how many cells are active and who belongs to each one.

Two codenames are in the same cell if one can be rearranged to form the other.

Input:

- First line: an integer n ($1 \leq n \leq 10^4$) — the number of codenames.

- Second line: n space-separated strings, each consisting of lowercase English letters (length ≤ 100).

Output: Print each group of codenames on a separate line (order of groups and within groups does not matter).

```
Input:
6
eat tea tan ate nat bat
```

```
Output:
eat tea ate
tan nat
bat
```

Question 4 — The War Council

A kingdom is preparing for war. The king has n generals, each with a different troop strength. Before marching, the king demands the generals line up in ascending order of strength. The royal advisor insists they use the ancient **Divide and Conquer** formation strategy:

1. Split the generals into two halves.
2. Recursively sort each half.
3. Merge the two sorted halves into one ordered line.

You must simulate this process. **No magical shortcuts (built-in sort functions) are allowed.**

Input:

- First line: an integer n ($1 \leq n \leq 10^5$) — the number of generals.
- Second line: n space-separated integers representing troop strengths ($-10^9 \leq \text{arr}[i] \leq 10^9$).

Output: Print the sorted array on a single line.

```
Input:
7
38 27 43 3 9 82 10
```

```
Output:
3 9 10 27 38 43 82
```

Expected complexity: $O(n \log n)$ time, $O(n)$ space.

Question 5 — The Tribal Council Election

A tribal council of n elders must elect their leaders. Each elder casts a vote (represented as an integer — the candidate's ID). The tribe's ancient law states that only candidates who receive **more than $\lfloor n/3 \rfloor$ votes** are declared leaders. There can be **at most two** such leaders.

Given the list of votes, find all candidates who received more than $\lfloor n/3 \rfloor$ votes.

Input:

- First line: an integer n ($1 \leq n \leq 10^5$) — the number of elders.
- Second line: n space-separated integers representing the votes cast.

Output: Print the winning candidate IDs in any order, separated by spaces. If no candidate qualifies, print `-1`.

Input:

6

1 1 1 3 3 2

Output:

1 3

Input:

7

1 2 3 4 5 6 7

Output:

-1

Input:

8

4 4 4 4 1 2 3 4

Output:

4

Constraints:

- You must solve it in **$O(n)$** time and **$O(1)$** extra space.
- Hint: Think about how many candidates can possibly exceed $\lfloor n/3 \rfloor$ votes, and try to narrow them down in a single pass.

Good luck, warrior!